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Design and Development of a VR System for Familiarizing Users with Collaborative Robot Safety Functions

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Abstract. One of the main challenges in introducing collaborative robots in industry is trust. A commonality in trust definitions in the areas of automation and robotics is the uncertainty associated with unfamiliar situations, where initial trust is influenced by knowledge and experience. For collaborative robots, many safety functions are not visible to users and therefore require knowledge or experience to understand. Becoming familiar with these functions can help users feel less uncertain and vulnerable when working with or near robots. Virtual reality can provide a safe and realistic environment where users can become familiar with new technology. To address this, we propose developing a virtual reality system that enables users to experience some of the safety functions of a collaborative robot. The system is being built in Unity, and the virtual environment includes a representation of an ABB GoFa robot. Through hands-on interactions, users can experience these safety functions and the robot's corresponding behavior when a safety limit is exceeded by using hand-guiding. These events are reinforced using sensory feedback to enhance the perceivability of the safety functions' limits and the robot's safety-related behaviors. For example, vibration intensity increases as the robot approaches its speed limit. The goal of the developed system is to explore whether virtual exposure to these safety functions can help establish initial trust in collaborative robots before real-world interaction, by increasing knowledge and experience.

1 Introduction

Collaborative robots for industrial purposes have been in use for many years and are well-suited for operations such as assembly, material handling, and pick-and-place [1]. Manufacturers market the robots as safe to operate without being confined to fenced cells. However, before this can be realized, safety standards must be taken into account to ensure physically safe operation, safe working close to and with human workers, and a risk assessment [2, 3]. When it comes to

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safety, some of the safety functions of collaborative robots are invisible, as they do not always require physical equipment. Examples include power and force limiting, monitored-speed, and software-based limiting [2, 3]. These often invisible functions can be viewed as affordances of collaborative robots and even as hidden affordances. Hidden affordances are opportunities for actions in an object that are not directly visible and require specific knowledge and experience to understand [4]. To make the hidden affordances of these types of functions more apparent to users, they can be complemented with signifiers. Signifiers are cues that make affordances apparent and can also reinforce the functionality itself [5].

A safe way for an operator to familiarize themselves with invisible safety functions, hidden affordances, is to use Virtual Reality (VR). Allowing operators to build knowledge and experience with safety functions is also a way to address one of the key challenges in introducing collaborative robots at the industrial shop-floor, namely trust [6, 7, 1, 8], which has gained growing attention in recent years [9, 10, 11]. VR is a cost-effective and safe tool to use since no physical industrial equipment is required [12]. It has been proposed as a key enabler for human-robot collaboration in the industry, both as an evaluation and training tool [13, 14]. In the context of industrial robots, an early focus of VR was primarily on task-specific operator training, along with evaluations of the virtual environment itself, focusing on aspects such as immersion and presence [15, 16, 17, 18]. In recent years, interest in using VR for subjective aspects, such as trust, has increased, including trust-building [10] and trust recovery [19] with collaborative robots for industrial purposes.

This paper presents the design and development of a VR system for familiarizing users with collaborative robot safety functions. Currently, there exist few systems tailor-made specifically to make safety functions (such as power and force limiting, monitored-speed, and software-based limiting) tangible to operators through interactive and sensory-rich experiences. The proposed system addresses this gap by enabling operators to explore how these safety functions operate and respond within a realistic, immersive environment.

By combining visual and vibrotactile feedback, the system provides perceivable cues that allow users to intuitively understand the safety boundaries and behavioral responses of collaborative robots, thereby supporting initial trust. From an industrial perspective, the proposed system can serve as a safe, cost-effective training and evaluation tool, preparing operators to work with collaborative robots under realistic and safe conditions. From a research perspective, the system enables studies of factors such as trust, familiarity, and perceived safety in controlled environments.

The remainder of this paper is structured as follows. Section 2 presents a theoretical background. Section 3 describes the system design. Section 4 includes the discussion, and Section 5 presents the conclusion.

2 Background

This chapter gives an overview of core concepts of the study.

2.1 Collaborative Robots

Collaborative robots are marketed by manufacturers as user-friendly, easy to program, and operate. These robots are applicable in various industrial areas, such as manual assembly and material handling [1]. Differences between them and traditional industrial robots include generally lower payload capabilities and adjustable characteristics [20]. A collaborative robot arm consists of rotational joints at each connection point. A mounting surface is located at the last rotational joint on the robot arm [2], to which an end-effector, or tool, can be attached. A common way to program and operate both industrial and collaborative robots is through a

teach pendant [3]. A teach pendant is a handheld device that allows programming, operating the robot, and activating and deactivating functions related to both program structure and safety functions. Robots use a point in space, usually at the tip of the tool, which reaches the added target or waypoints. This point is referred to as a Tool Center Point (TCP) [2].

Collaborative robots are equipped with safety functions defined in safety standards, and some are mandatory, conditional, or optional depending on the application [2, 3]. These functions ensure the physical safety of humans working closely with or alongside robots. A method for controlling a robot's movements hands-on is hand-guided control [2]. To use this method, certain safety functions must be activated, such as hold-to-run control, monitored-speed, monitored-standstill, and software-based limiting. Hold-to-run control allows the robot to move only as long as an enabling device remains active. Monitored-speed continuously monitors the robot's Tool Center Point (TCP) speed. Monitored-standstill monitors the robot to ensure that there is no unintended motion. Software-based limiting restricts the robot's movements to a specific area.

2.2 Virtual Reality

VR is an environment where everything included is virtual, and where an experience is total devotion to the simulated environment [21]. Essential aspects to consider when it comes to virtual environments include immersion, presence, and sensory fidelity.

Immersion is an objective aspect, referring to being included in a virtual environment [22], instead of just observing it. The level of immersion is dependent on the technology used and the system design. Experiencing VR with a head-mounted display, described as immersive VR, enables users to engage more senses compared to experiencing VR on a computer screen, described as non-immersive VR [13]. In contrast, presence is a subjective aspect and refers to the feeling of being present in a virtual environment [22].

Another aspect related to VR is sensory fidelity, which refers to the realism of the virtual environment and its sensory cues [23]. It is preferred to activate multiple senses within a virtual environment, especially a combination of visual and vibrotactile feedback, engaging both visual and tactile senses, as this can enhance the perceived realism within a virtual environment [24].

Sensory feedback is used to engage users' senses, such as visual and vibrotactile feedback [23]. Visual feedback is often used to indicate to users when and where to grasp and release objects within the environment [25], usually related to color effects and changes. Another way to indicate and instruct users is to use text labels [26]. Vibrotactile feedback, a type of tactile feedback that allows users to feel vibrations on their skin, typically from the VR controllers, is often used to notify users of events occurring within a virtual environment [27].

2.3 Measuring Trust

Various trust definitions are used in the literature [28, 29, 30, 31], some of which originate from the field of automation and were later adopted in the field of robotics [31]. Unfamiliar situations and transfer of control are common features of various trust definitions [32]. Trust that exists before an interaction has occurred is referred to as initial trust and is influenced by factors such as existing knowledge and prior experience [28, 33, 34].

Self-reporting-based measurement scales are a common way to measure trust [35, 30, 36]. Some of the trust-related factors and statements included in these scales are context- and/or task-dependent, meaning that some, especially those that are task-dependent, can be difficult to rate before an interaction. However, two of the measurement scales provide guidance for measuring initial trust [30, 36], focusing on trust factors and statements which can characterize trust level before an interaction has occurred.

One of these scales divides its trust factors into two dimensions, performance and moral [30, 37], with the moral dimension being more closely related to interactions in social robotics. The other scale has an alternative compact version focused more on performance-related aspects [36, 38]. The full scale also includes aspects more related to interactions in social robotics. Since completing a physical task is often central to collaborative robots in industrial contexts, it suggests that performance-based factors are more suitable for collaborative robots in these contexts compared to factors covering moral and ethical considerations. Results from a previous study regarding identifying applicable trust-related factors for collaborative robots [39] show that the factors included in the performance-based dimension [30] and the compact version [36, 38] align. This suggests that these scales are a suitable alternative for measuring initial trust in collaborative robots in industrial contexts.

3 System Design

This chapter presents the design of the proposed system and its functionality.

3.1 Overview of the System

Collaborative robots have many opportunities for action that humans can perceive, affordances, however, some of these are invisible, requiring specific knowledge or exploration to understand. These hidden affordances can be complemented with signifiers in VR to make them more understandable to users, making them perceivable by reinforcing their intended function without the need for physical equipment. The main design goal of the virtual system is to create a realistic and representative environment that closely mimics the functionality and corresponding behavior of real collaborative robots' and their safety functions. This enables users to enhance their knowledge of and experience with these robots on a more general level.

The included safety functions in the virtual environment are power and force limiting, monitored-speed, and software-based limiting. These three functions can be experienced and tested hands-on through the hand-guided control function.

In practice, to use the hand-guided control function, certain safety functions must be active according to the standards, including monitored speed, software-based limiting, hold-to-run control, and monitored-standstill. The hold-to-run control function activates the hand-guided control and remains active as long as the hold-to-run control is active. Monitored-standstill will not be included in the virtual system.

3.2 Design Considerations: Immersion, Fidelity and Presence

Immersion, sensory fidelity, and presence are considered essential aspects when designing VR experiences. To achieve immersion, the system should surround the user by activating multiple senses and enabling interaction with the environment rather than passively observing it. Sensory fidelity, on the other hand, is related to the level of realism in the environment and its sensory cues. To achieve presence, the system should enable users to feel present in the environment by interactivity, activating multiple senses, and including realistic representations of both the environment and its sensory cues.

Informed by these concepts, the virtual system is designed to deliver a realistic, interactive experience. The system uses a head-mounted display and controllers to enable users to experience and interact with the environment. Both visual and vibrotactile feedback are used to activate multiple senses. By activating multiple senses, users will not only observe how these safety functions work but also experience them physically, aligning with the concept of immersion.

The robot's behavior and the interactions inside the environment are similar to reality, aligning with the concept of sensory fidelity. Additionally, both hold-to-run control and hand-guided control are included to replicate physical interactions with a collaborative robot.

Presence, on the other hand, is a subjective experience that may vary between participants. It is supported by interactive elements within the virtual environment, realistic representations of the environment and tasks, and sensory feedback.

3.3 Virtual System

The system is developed and implemented in Unity 3D. A Meta Quest Pro headset is used to let participants experience the environment, and the two hand controllers enable interactions within it (figure 1). The controllers are used to grasp the robot, hand-guide it, and push virtual buttons.

A virtual representation of a collaborative robot is located inside the environment, and its motions are simulated using a custom script. The robot is only allowed to move linearly during hand-guiding, meaning that its TCP can only move in the X, Y, and Z directions, thereby restricting the TCP from rotating. The robots' allowed working zone is restricted to ensure the movements mimic the real robot's motion pattern.



Figure 1. A Meta Quest Pro headset and controllers.

3.4 Sensory Feedback for User Guidance

In the virtual environment, visual feedback guides users' actions by indicating what and how to interact with objects and informing them about the specific safety functions in the various scenarios. This includes color changes, halos around objects, and textual instructions.

Color changes are applied to existing objects when users collide with them with the controllers. Halos are used in the environment to indicate to users when an object can be interacted with (figure 2). The textual instructions are both task-dependent and informative. These instructions update dynamically as users progress, meaning that not all instructions will be visible at the same time.



Figure 2. In the virtual environment, the grasp positions, including the hold-to-run control, used to activate the robot for hand-guiding, are highlighted with halos as visual feedback.

3.5 Reinforcing Sensory Feedback of the Safety Functions

Power and force limiting is reinforced with visual feedback by adding a light indicator, positioned on the last link of the robot, as a pressure-based color indicator. This indicator will represent the pressure that the last joint of the robot experiences when it comes into contact with another object. The color indicator will transition from white, indicating no contact to very low pressure, to green when the applied pressure is within the preferred range and starting to approach the limit, to yellow as the pressure begins to reach the limit, and to red when it has exceeded the maximum limit, causing the robot to enter a protective stop. Power and force limiting will also use vibrotactile feedback that increases in intensity as the pressure increases, following an exponential curve to emphasize the approach toward the maximum limit.

Monitored-speed limiting is reinforced with visual feedback by adding a light indicator, positioned on the last link of the robot, as a speed-based color indicator that represents the TCP speed. The color indicator will transition from white to green as the TCP starts to move, until it reaches approximately 80 percent of the allowed speed limit. It will then transition to yellow when it is close to the limit, and finally to red when it exceeds the limit, causing the robot to enter a protective stop. Monitored-speed will also use vibrotactile feedback that increases in intensity as the speed increases, following an exponential curve to emphasize the approach toward the maximum limit.

Software-based limiting is reinforced with visual feedback by adding a light indicator positioned on the last link of the robot, whose color transitions are based on the robot's distance from the allowed working zone. The color indicator starts green in its initial position, located within the working zone, and transitions to yellow as the robot's TCP approaches one of the edges. It finally turns red when it exceeds the allowed working zone, causing the robot to enter a protective stop. Vibrotactile feedback will also be used, which increases in intensity as the robot's TCP approaches one of the edges, following an exponential curve to emphasize the approach toward the edges of the allowed working zone.

The reinforcing visual feedback consisting of light indications could potentially be added in real applications, and two of the three safety functions include vibrotactile feedback that

corresponds to the robot's inertia during increased movement and the force experienced when applying pressure to a surface.

3.6 System Sequence

The virtual experience begins with a training scenario where users will learn how to interact with the virtual environment using the controllers. All instructions and general guidance in the environment are provided through visual feedback and are consistent across all scenarios.

The following three scenarios are presented in random order, and each includes an introduction to the corresponding safety function and its reinforcing signifiers, followed by a task in which that function and its signifiers are active. In the introduction, users must trigger the safety function's limit at least a couple of times before they can proceed to the task. Users are encouraged to complete the task as fast as possible without exceeding the safety limit.

After all the functions are introduced and used within their respective task, a workflow that combines all previously performed tasks, with their specific safety functions active, will be performed.

3.7 Workflow Structure

The safety functions, power and force limiting, monitored-speed limiting, and software-based limiting in the virtual system are used together with the hand-guided control function. To activate the robot and enable hand-guided control, the hold-to-run control function must be activated. To activate it, the buttons on the handlebars, one on each side of the robot's last link, must be pressed and held down, enabling the robot to move (figure 3). Each safety function will be active during a specific task performed by the participants. These tasks will then be combined at the end of the experiment and executed as a workflow.

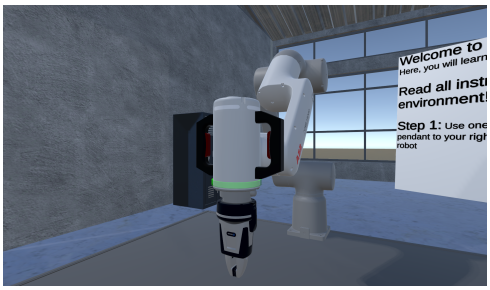


Figure 3. The robot in the virtual environment, the handle located on each side of the robot acts as the hold-to-run control, and the green color indicator located on the robot's last link acts as the visual feedback indicator.

3.8 Usage and Evaluation Context

The virtual system is designed to educate future users about safety functions of collaborative robots in industrial contexts and to explore its effect on initial trust. A previous study identified applicable trust-related factors for collaborative robots in [39], which were based on two existing measurement scales of trust in robots. These measurement scales were the performance-based part of the Multi-Dimensional Measure of Trust (MDMT) [30, 37] and the shorter 14-item version of the Trust Perception Scale-HRI (TPS-HRI) [36, 38]. Both scales mainly focus on performance-based aspects of trust, which aligns with the focus of collaborative robots in industrial contexts. The mapping between the evaluated trust-related factors and the two existing

measurement scales as well as explanations of the factors' meanings can be seen in Table 1. Safety, trustworthiness, and familiarity were included in the evaluation in [39]. However, they are not included in the table, as they were considered overarching factors that reflect general perceptions of collaborative robots in industrial contexts. Even if not all factors have an exact corresponding factor in the scales, the included factors were chosen to broadly represent the trust-related factors and statements included in both scales.

One or both of the identified trust scales will be used to measure initial trust. Initial trust will be measured both before and after using the virtual system.

Table 1. Mapping between evaluated factors in [39] and items from the MDMT [30, 37] and TPS-HRI [36, 38] scales.

Factor	MT (MDMT)	TPS-HRI (14-items)	Explanation
Reliability	Reliable	Reliable / (Malfunction) / Dependable	The extent to which the robot is stable and consistent in fulfilling tasks
Predictability	Predictable	Predictable / (Unresponsive) / Act consistently	How easily the robot's behavior can be anticipated in different situations
Competence	Competent	Function successfully / Meet the needs of the mission/task	The degree to which the robot is capable of completing its assigned tasks effectively
Usability	Competent	Function successfully / Meet the needs of the mission/task	The degree to which the robot is easy and effective to use
Capability	Capable	Perform exactly as instructed / Function successfully	The extent to which the robot's abilities are perceived to handle various tasks
Adaptability	Capable	Function successfully / Meet the needs of the mission/task	How flexibly the robot can adjust to different work conditions
Accuracy	Consistent / Meticulous	(Have errors) / Act consistently	The extent to which the robot performs tasks with precision and minimal errors
Appropriateness	Competent / Skilled	Meet the needs of the mission/task	How suitable the robot's behaviors and functions are for manual assembly
Intuitiveness	Predictable	Communicate with people / Provide feedback / Provide appropriate information	The extent to which the robot is easy to understand or interact with

4 Discussion

The developed VR system demonstrates how immersive technologies can support users' understanding of collaborative robot safety functions. The implementation of multimodal feedback, combining visual and vibrotactile cues, makes the robot's safety behavior more intuitive. A consistent relationship between user actions, sensory feedback, and robot responses enables users to quickly interpret the system's boundaries and safety conditions. The idea is that well-designed sensory mapping can strengthen users' mental models and improve confidence when interacting with collaborative robots.

The scenario-based structure of the system, in which each safety function is introduced and demonstrated before being combined in a full workflow, is meant to provide an effective approach to gradual learning and skill transfer. It supports progressive familiarization and could facilitate retention of both procedural and conceptual knowledge. While we believe that the developed systems is promising, the degree to which the virtual experiences lead to measurable improvements in trust and safe behavior in real-world human-robot interaction remains to be systematically evaluated. Controlled studies using validated trust scales and behavioral performance metrics will be undertaken to confirm the system's effects.

Certain limitations of the current prototype should be mentioned. The system presently models only a subset of the robot's safety functions and simplifies the robot's behavior. Expanding the model to include more complex motion patterns, physical dynamics, and real-time response would increase realism and applicability. Moreover, the lack of empirical evaluation limits conclusions about the effectiveness of the multimodal feedback, but this is something that will be investigated further on.

5 Conclusions

The prototype system developed shows technical feasibility and strong potential for both training and research purposes. The system can function as a safe, repeatable, and cost-efficient tool for operator familiarization, enabling exploration of robot safety behavior without risk. It also provides a flexible experimental platform for investigating human–robot collaboration, where parameters such as feedback type or motion profiles can be varied systematically. The results highlight the value of using VR not only as a visualization medium but also as a controlled environment for analyzing trust, perception, and understanding of robotic systems.

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